



Why do microplastics aggravate cholestatic liver disease? The NLRP3-mediated intestinal barrier integrity damage matter[☆]

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ABSTRACT

Microplastics (MPs) are becoming a significant environmental and public health concern because they are present in freshwater and marine environments and are ingested by living organisms. Cholestatic liver disease (CLD) is closely related to intestinal homeostasis, but there are no data investigating the effects of MPs on CLD. In this study, we used *Mdr2*^{-/-} mice (a model of CLD) to investigate the effects of polystyrene microplastics (PS-MPs, 0.5 μ m) on CLD and the underlying mechanisms. Our data revealed that, compared with *Mdr2*^{-/-} mice, PS-MPs (200 μ g/day)-challenged *Mdr2*^{-/-} mice presented more severe collagen deposition, infiltration of inflammatory cells in liver sections and higher alkaline phosphatase (ALP)/ γ -glutamyltransferase (γ -GGT) concentrations in the serum. Furthermore, the number of mucous cells in the colonic tissues of mice with CLD was strongly inhibited by PS-MPs, accompanied by the downregulation of intestinal barrier integrity proteins (ZO-1, Occludin and Claudin-1). Through correlation analysis to further verify the connection between ALP/ γ -GGT levels and intestinal barrier integrity genes, as well as a significant positive correlation with IL-1 β after PS-MPs exposure. Our results also revealed that PS-MPs exposure accelerated the NOD-like receptor protein 3 (NLRP3)-associated inflammatory response in the colon but did not affect NLRP3 expression in the livers of *Mdr2*^{-/-} mice. Further study confirmed that the inhibition of NLRP3 by the MCC950 inhibitor abrogated the exacerbating effects of PS-MPs on hepatobiliary injury and intestinal barrier integrity damage. These findings provide the first evidence that NLRP3-mediated inflammation is an important participant in intestinal barrier integrity damage crosstalk that drives CLD under MPs exposure and identify NLRP3 as a potential therapeutic target.

1. Introduction

Plastic products are widely used in various fields, and owing to their stable chemical structure, they can be transferred and accumulate in nature for a long time and do not degrade easily. Since the beginning of the COVID-19 pandemic, the massive use of disposable products such as masks has further exacerbated the environmental burden, and plastic pollution has become an emerging global environmental problem (Song et al., 2024). In 2019, 460 million tons of plastic were consumed globally, but only 55 million tons were properly recycled (Oecd, 2022). A study in the state of Kochi reported an average of 21.27 particles $m^{-2} d^{-1}$ in atmospheric fallout (Kannankai, 2024). On the Tibetan Plateau, China, the average concentration of microplastics in lake surface water is 5.56 items/L (Liang et al., 2024). In southern China, the average amount of microplastics in the surface water of eleven investigated

nature reserves and scenery districts ranged from 542 to 5500 items/ m^3 (Gong et al., 2023). Plastics have become a serious global environmental challenge because of their low degradability, ease of transport and accumulation, and ecotoxicity (Yin et al., 2023).

Microplastics (MPs) (diameter <5 mm) are frequently detected in human organs such as the liver (Horvatits et al., 2022), lung (Amato-Lourenco et al., 2021), placenta (Ragusa et al., 2021) and blood (Leslie et al., 2022), with polystyrene microplastics (PS-MPs) being among the most common types of MPs found in coastal environments, surface waters and soils. Humans are exposed to an abundance of PS-MPs through various routes, with oral ingestion considered to be the main route (Lett et al., 2021). Oral PS-MPs can lead to the destruction of intestinal epithelial cells, inflammation of the intestinal wall and decreased intestinal mucus volume (Huang et al., 2023; Huang et al., 2021). Furthermore, PS-MPs exposure results in dysbiosis of the

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intestinal flora and disturbed gut metabolism in zebrafish (Jin et al., 2018). Notably, the concentration of MPs in the feces of patients with inflammatory bowel disease (IBD) is significantly greater than that in the feces of healthy people (Wang et al., 2023). Furthermore, the concentration of MPs in feces is positively correlated with the severity of disease, suggesting that MPs may intervene in the intestinal microecology, thus increasing the susceptibility of organisms to other diseases (Yan et al., 2022). For example, a study on IBD revealed that PS-MPs intervention exacerbated intestinal histopathological damage and intestinal permeability destruction, subsequently resulting in secondary liver injury in colitis model mice (Luo et al., 2022). Therefore, the potential adverse effects of PS-MPs on the risk of sensitive populations should not be ignored, and relevant research is urgently needed.

Cholestatic liver disease (CLD) is a general term for a group of hepatobiliary diseases caused by environmental factors that result in impaired bile formation, secretion and excretion and thus bile acid stagnation, leading to liver lesions. The pathogenesis of CLD is complex, with an insidious onset. In the progressive stage, hyperbilirubinemia, such as jaundice and pruritus, may occur, eventually progressing to hepatic fibrosis, cirrhosis and liver failure (Li et al., 2017). A close association with gut inflammation is an important feature of CLD, and approximately 90% of patients experience gut inflammation (Palmela et al., 2018). Compared with healthy individuals, CLD patients have unique gut microbiological characteristics, with studies showing aberrant proliferation and senescence of cholangiocytes, suggesting the importance of cholangiocytes and the adaptive immune response in the course of CLD (Kummen et al., 2017). The "leaky gut" hypothesis suggests that NLRP3-associated intestinal inflammation may lead to intestinal barrier injury, subsequently changing gut microbial species and abundance and breaching hepatic immune tolerance mechanisms, thus driving the development of hepatobiliary disease (Giordano et al., 2018). Additionally, NLRP3 activation promotes mutual crosstalk between hepatocytes and macrophages to exacerbate bile acid-induced hepatocyte inflammation (Song et al., 2022). NLRP3 acts as a scaffold, recruiting apoptosis-associated speck-like proteins (ASCs) to interact with pro-caspase-1, subsequently activating the precursors of pro-IL-1 β into active maturation factors (Qu et al., 2019). Furthermore, Caspase-1 leads to the cleavage of GSDMD and the release of the N-terminal domain, which then binds to phospholipids on the cell membrane and exhibits functional pore activity, which is conducive to the release of IL-1 β , thus triggering an inflammatory response (Li et al., 2023; Shi et al., 2024; Yuan et al., 2022). However, the role of NLRP3 inflammasome-intestinal barrier crosstalk in the effects of MPs on the course of CLD *in vivo* remains unclear.

Given the limited efficacy and poor prognosis of CLD, it is necessary to evaluate the risk of exposure to PS-MPs in susceptible populations, which will provide an experimental basis for the formulation of relevant environmental health standards, elucidate the possible pathogenesis of CLD, and enrich the ideas for clinical intervention and treatment. The purpose of this study was to investigate the effects of PS-MPs on CLD and explore the possible mechanism of PS-MPs in the development of CLD. Herein, Mdr2^{-/-} mice exposed to 0.5- μ m PS-MPs presented more severe hepatobiliary injury than did Mdr2^{+/+} mice. Additionally, PS-MPs exposure resulted in colonitis and intestinal barrier integrity damage and accelerated CLD in Mdr2^{-/-} mice. Furthermore, mouse experiments revealed that NLRP3 ablation protected against PS-MPs-induced aggravation of CLD. Our findings firstly demonstrated that NLRP3/intestinal barrier integrity damage axis are involved in PS-MPs aggravate CLD, which suggests a possible treatment approach for PS-MPs-induced early stages of hepatic diseases.

2. Materials and methods

2.1. Chemicals and reagents

Polystyrene microplastics (PS-MPs, 0.5 μ m) suspensions were

purchased from the Baseline Chrom Tech Research Centre (Tianjin, China). The characterization of the microplastics is detailed in [Supplementary Fig. S1](#). The selective NLRP3 inflammatory inhibitor MCC950 (256373-96-3) was purchased from Bidepharm (Shanghai, China). MEM containing NEAAs was purchased from Pricella (Wuhan, China), and bovine serum albumin (BSA) was obtained from Solabio Scientific (Beijing, China). A mouse tail direct PCR kit was purchased from Foregene (Chengdu, China). Commercial assay kits (Nanjing, China) were used to determine serum biochemical index activities. Trypsin, TRIzol, reverse transcription, and dye-based colour fluorescence quantitative PCR premix kits were obtained from Titan (Shanghai, China). Radio-immunoprecipitation assay (RIPA) strong lysis buffer, a BCA protein concentration assay kit, 5 $\times$ SDS-PAGE protein sampling buffer and an anti-fluorescence burst sealer (with DAPI) were purchased from Beyotime Biotechnology (Shanghai, China). Polyvinylidene fluoride (PVDF) membranes (0.45 μ m) were purchased from Millipore (Millipore, USA). The superior ECL luminescent liquid was obtained from ShareBio (Shanghai, China).

2.2. Animals and grouping

Mdr2^{-/-} mice (Friend Virus B NIH Jackson (FVB/NJ) background) were purchased from Jackson Laboratory in the United States. The mice were amplified by heterozygous pair-breeding according to Mendelian laws of inheritance. According to the animal genotyping program of Jackson's laboratory, the wild type presented a band at only 202 bp, the heterozygous knockout individuals (Mdr2^{+/-}) presented bands at both 202 bp and 272 bp, and the mutant (Mdr2^{-/-}) presented a band at only 272 bp ([Fig. 1A](#)). All animal experiments were approved by the Laboratory Animal Welfare Ethical Committee (XYGW-2021-93). The ethical certification of laboratory animals is shown in [Supplementary Fig. S2](#). The DNA template solution of mouse tail tissue was obtained by digesting and lysing the tissue, followed by incubation at 65 °C for 10–30 min, inactivation at 95 °C for 5 min, and centrifugation. This solution was then used for PCR with an identification primer sequence synthesized by Sangon Biotech (Shanghai, China). Mdr2^{-/-} and wild-type (WT) mice shared the same forward primer: 5'-TGGGAA-GAGTGGAGAAATCG-3', the reverse primer of Mdr2^{-/-}: 5'-GCTACTTCCATTGTTCACGTCC-3', and the reverse primer of WT: 5'-TGAAGACATCGGTGTTTCAGAG-3'. After agarose gel electrophoresis analysis, individuals with only a 272 bp band were identified as homozygous mice for subsequent experiments ([Fig. 1A](#)). The homozygous mice that were identified were kept in animal facilities at 24 $\pm$ 0.5 °C with 50–65% humidity and a standard 12-h dark/light cycle. All animal operations in the study were conducted in accordance with the ethical standards for laboratory animal welfare of Central South University and were based on the guidelines of the National Institutes of Health.

To evaluate the effects of PS-MPs on Mdr2^{-/-} mice, 8-week-old male Mdr2^{-/-} mice were randomly divided into four groups (n = 10 per group): the Mdr2^{-/-} mouse group, the Mdr2^{-/-} + PS-MPs group, the Mdr2^{-/-} + PS-MPs + MCC950 group and the Mdr2^{-/-} + MCC950 group. In the Mdr2^{-/-} + PS-MPs group, Mdr2^{-/-} mice received PS-MPs (0.5 μ m) by gavage at a concentration of 200 μ g/day for 28 days, and MCC950 was administered by intraperitoneal injection at a dose of 10 mg/kg on days 0, 1, 2, and 4 and every two days thereafter. The dose and duration of PS-MPs exposure to Mdr2^{-/-} mice were described in a previous study (Shi et al., 2022a). At the end of the experiment, male mice were fasted overnight and then sacrificed. The serum, liver, and colon samples were collected and immediately flash-frozen in liquid nitrogen before being stored at -80 °C until use.

2.3. Cell culture

Caco-2 cells were purchased from Pricella (Wuhan, China) and cultured in MEM supplemented with NEAAs, 20% fetal bovine serum and 1% penicillin-streptomycin. The cells were incubated in an

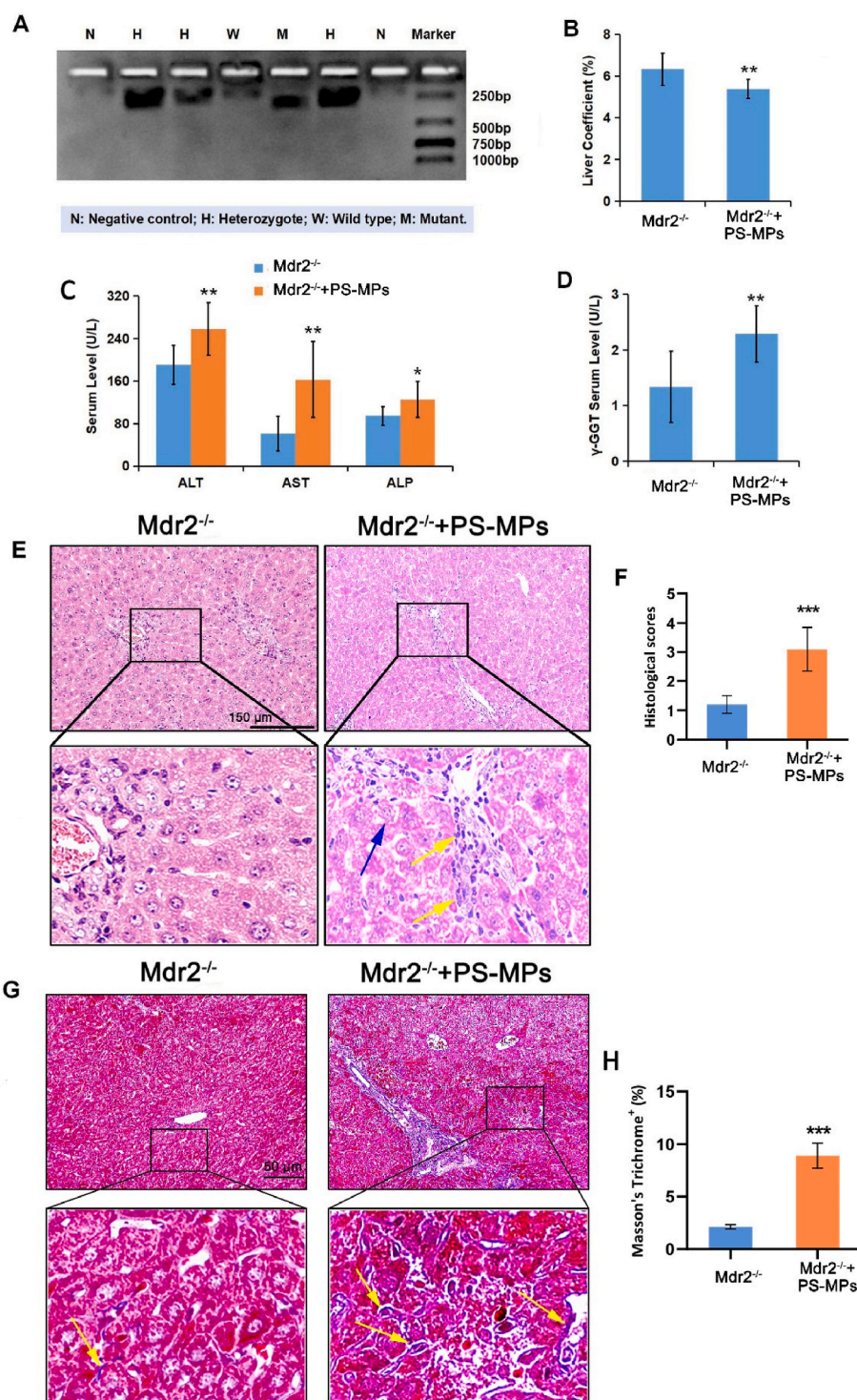


Fig. 1. Effects of PS-MPs on the growth phenotype and liver injury of Mdr2^{-/-} mice. (A) Identification of the genotypes of Mdr2^{-/-} mice. (B) Liver coefficient. (C) Serum levels of ALT, AST, and ALP. (D) Serum γ-GGT levels. (E) H&E staining of liver sections. Inflammatory cell infiltration is shown by yellow arrows, and mild vacuolar degeneration of hepatocytes is shown by blue arrows. (F) Representative liver histological scores (H&E). (G) Masson staining of liver sections. (H) Quantification of Masson's trichrome staining. n = 10 mice per group. Significant correlations are represented by **P* < 0.05, ***P* < 0.01, ****P* < 0.01 vs. the Mdr2^{-/-} group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

incubator at 37 °C with a CO₂ concentration of 5%. When the confluence reached 80–90%, the cells were digested with trypsin and subjected to passaging treatment. The control, PS-MPs, PS-MPs + MCC950 and MCC950 groups were established. In the PS-MPs group, Caco-2 cells were treated with PS-MPs with a diameter of 0.5 μm at a concentration of 120 μg/mL for 24 h or 48 h, and Caco-2 cells were pretreated with the

selective NLRP3 inflammasome inhibitor MCC950 at a concentration of 10 μM 1 h before the formal experiment until the end of the entire exposure process.

2.4. Histopathological evaluation

At least 3 mice in each group were randomly selected, and their fresh liver tissues and colon tissues were fixed with 4% paraformaldehyde. After 24 h of fixation, the tissues were dehydrated, hyalinized, embedded in paraffin and cut into 5 μ m thick slices. The liver slices were stained with hematoxylin-eosin (H&E) and Masson's trichrome, and the colon tissue slices were stained with H&E and alcian blue/periodic acid Schiff (AB-PAS) and then observed under a microscope. Histopathological scores for hepatic and colonic damage were evaluated by certified pathologists who were blinded to the study following the criteria described previously (Supplementary Table 1S) (Ai et al., 2022).

2.5. Determination of serum biochemical parameters

The activities of alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP) and γ -glutamyltransferase (γ -GGT) in the serum were determined via commercial kits.

2.6. Real-time quantitative polymerase chain reaction (RT-qPCR)

Total RNA from colon tissues and Caco-2 cells was fully cleaved and extracted with TRIzol reagent as described in our previous study (Li et al., 2021a). After cDNA was synthesized with a one-step reverse transcription kit, the primer sequences of related genes were synthesized by Sangon Biotech (Shanghai, China). The mRNA levels were measured via a dye-based color fluorescent quantitative PCR premix kit to evaluate the mRNA levels of various indicators. The PCR thermocycling regimen was 95 $^{\circ}$ C for 5 min, followed by 40 cycles of 95 $^{\circ}$ C for 10 s and 60 $^{\circ}$ C for 30 s. Fold changes in target mRNA normalized to β -actin were calculated via the $2^{-\Delta\Delta C_t}$ method. The primer sequences of the genes are shown in Supplementary Table 2S.

2.7. Western blotting

Colon tissues, liver tissues and Caco-2 cells were lysed with RIPA strong lysis buffer containing 1 mM PMSF for 60 min, and the supernatant was extracted to obtain protein from the tissues and cells after centrifugation for 15 min at 4 $^{\circ}$ C and 15000 $\times$ g, as described in our previous study (Li et al., 2021b). The protein concentration was measured by using an enhanced BCA protein concentration determination kit. The concentration was adjusted by using PBS buffer, and then 5 $\times$ SDS-PAGE protein loading buffer was added and mixed. The protein was fully denatured in a water bath at 100 $^{\circ}$ C for 8–10 min and then stored at -80° C until use. Electrophoresis was performed on 10% and 12% SDS-PAGE separator gels and 5% SDS-PAGE concentrate gels to separate proteins from the samples, and the separated proteins were transferred to PVDF membranes in Tris-glycine buffer. The PVDF membrane was incubated with the primary antibody at 4 $^{\circ}$ C overnight, washed 3 times with TBST, and then incubated with the secondary antibody at room temperature for 1 h. The protein bands were visualized via Lumi Best ECL luminescent liquid, and the intensity values were quantitatively analysed via ImageJ software.

The antibodies used in this study were as follows: β -actin (K101527p) was obtained from Solabio Life Sciences (Beijing, China). ZO-1 (21773-1-AP), Occludin (27260-1-AP), Claudin-1 (28674-1-AP), MMP-9 (10375-2-AP), and GSDMD (20770-1-AP) were obtained from Proteintech Group (Wuhan, China). Caspase-1 (A0964), HRP-conjugated goat anti-rabbit IgG (H + L) (AS014), and HRP-conjugated goat anti-mouse IgG (H + L) (AS003) were purchased from ABclonal (Wuhan, China).

2.8. Immunofluorescence (IF) staining

After the Caco-2 cell culture medium was removed, the cells were washed 3 times with PBS, fixed with 4% paraformaldehyde, and

sequentially permeabilized and blocked. Caco-2 cells were incubated with primary antibody at 4 $^{\circ}$ C overnight and washed with PBS. The slides were subsequently incubated with a fluorescent secondary antibody dye for 2 h, followed by sealing with an anti-fluorescent burst sealing solution. The nuclei of the cells were stained with DAPI and visualized under a fluorescence microscope.

2.9. Statistical analysis

All data are expressed as the mean $\pm$ standard error of the mean (SEM), and statistical analysis and visualization were performed via SPSS 26.0 software and GraphPad Prism 9.0. Differences between two groups were compared via two-way t tests, differences among multiple groups were compared via one-way analysis of variance (ANOVA), and pairwise comparisons of multiple groups were compared via the least significant difference (LSD-t) test. Besides, Pearson correlation analysis is used to verify the correlation between genes. $P < 0.05$ was considered to indicate statistical significance.

3. Results

3.1. PS-MPs exposure exacerbated liver injury in *Mdr2*^{-/-} mice

We evaluated the potential effects of PS-MPs on cholestatic hepatitis in an *in vivo* model, which was achieved by daily gavage of PS-MPs in *Mdr2*^{-/-} mice. *Mdr2*^{-/-} mice exposed to PS-MPs presented a decreased liver coefficient compared with *Mdr2*^{-/-} mice (Fig. 1B). Moreover, the levels of serum ALT and AST and the activities of serum ALP and γ -GGT were greater in the *Mdr2*^{-/-} mice than in the *Mdr2*^{-/-} mice after PS-MPs exposure (Fig. 1C–D). H&E-stained sections revealed that PS-MPs exposure further promoted the infiltration of granulocytes and lymphocytes (yellow arrow) and aggravated connective tissue around the portal are (blue arrow) of *Mdr2*^{-/-} mice (Fig. 1E–F). Moreover, the number of blue-stained collagen fibres in the liver was greater in *Mdr2*^{-/-} mice exposed to PS-MPs than in *Mdr2*^{-/-} mice (Fig. 1G–H). Taken together, these findings suggest that PS-MPs exposure worsened liver fibrosis and inflammatory response in *Mdr2*^{-/-} mice.

3.2. PS-MPs exposure disrupted intestinal barrier integrity in *Mdr2*^{-/-} mice

To further explore the mechanism by which PS exacerbates the development of CLD, the extent of intestinal injury was examined. Compared with *Mdr2*^{-/-} mice, *Mdr2*^{-/-} mice exposed to PS-MPs presented decreased colon length (Fig. 2A–B). H&E-stained colon sections revealed that the structure of each layer of the colon tissue in *Mdr2*^{-/-} mice was clear, the mucosal epithelium was intact, the morphology of the epithelial cells was normal, and the mucosal layer lacked any obvious abnormalities. In contrast, after PS-MPs exposure, the lamina propria of colon tissue from *Mdr2*^{-/-} mice presented occasional loss of intestinal glandular structure, and many lymphocytes were focally distributed (blue arrow) (Fig. 2C–D). AB-PAS staining of the colon revealed that, compared with *Mdr2*^{-/-} mice, *Mdr2*^{-/-} mice exposed to PS-MPs presented a reduction in the number of goblet cells and focal intestinal glands in the lamina propria but an intact mucosal epithelium with abundant folds (black arrow) and a lower abundance of blue or bluish-purple goblet cells (green arrow) (Fig. 2E–F). Additionally, western blotting revealed that the protein expression of proinflammatory cytokines (IL-1 β , TNF- α and IL-6) in the colon was greater in PS-MPs-challenged *Mdr2*^{-/-} mice than in *Mdr2*^{-/-} mice (Fig. 2G). Furthermore, compared with *Mdr2*^{-/-} mice, *Mdr2*^{-/-} mice exposed to PS-MPs presented decreased protein levels of ZO-1, Occludin, and Claudin-1 and increased protein levels of O00(MMP-9) in the colon (Fig. 2H). Taken together, these results suggest that PS-MPs aggravate cholestatic liver injury by disrupting the intestinal barrier and increasing intestinal inflammation.

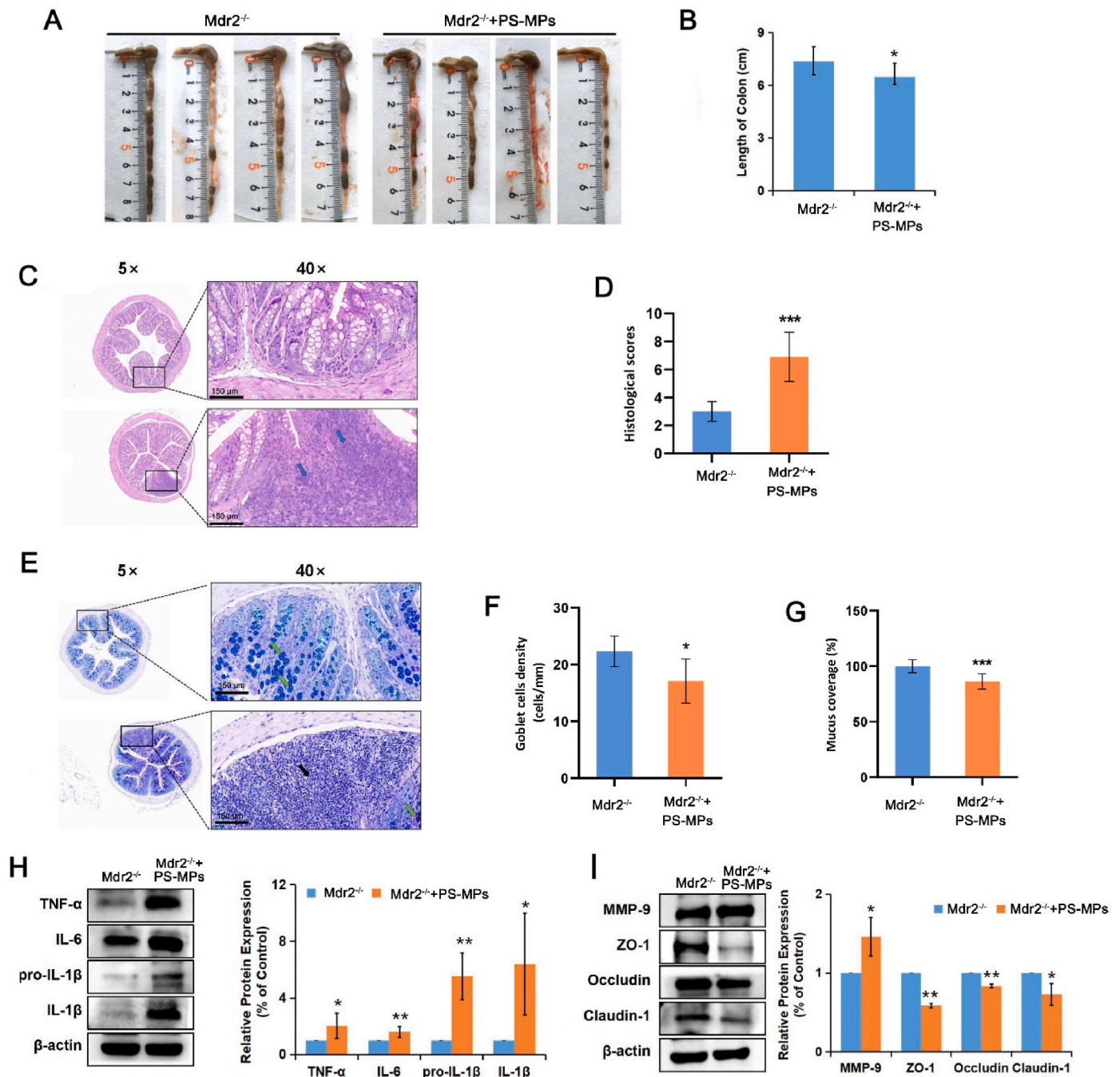


Fig. 2. Effects of PS-MPs on the colon tissue of $Mdr2^{-/-}$ mice. (A–B) Colon length and morphology of $Mdr2^{-/-}$ mice. (C) H&E staining of colon sections. Inflammatory cell infiltration is shown by blue arrows. (D) Representative colon histological scores (H&E). (E) AB–PAS staining of colon sections. Goblet cells are shown by green arrows, and the focal decreased number of lamina propria intestinal glands is shown by black arrows. (F) The corresponding goblet cell densities and (G) mucus coverage. (H) The protein expression levels of TNF- α , IL-6, pro-IL-1 β and IL-1 β in colon tissue were determined by western blotting and quantified. (I) The protein expression levels of MMP-9, ZO-1, Occludin and Claudin-1 in colon tissue were determined by western blotting and quantified. $n = 10$ mice per group. Significant correlations are represented by * $P < 0.05$, ** $P < 0.01$, *** $P < 0.01$ vs. the $Mdr2^{-/-}$ group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.3. NLRP3 inflammasome is involved in PS-MPs-induced colon inflammation in $Mdr2^{-/-}$ mice and Caco-2 cells

NLRP3 inflammasome pathway is mainly responsible for the activation and release of a cascade of proinflammatory mediators that contribute to the development of CLD. Next, we investigated whether NLRP3-driven colonic inflammation is involved in PS-MPs-induced aggravation of cholestatic liver injury. We found that there was no

difference in the protein of NLRP3 levels between PS-MPs + $Mdr2^{-/-}$ mice and $Mdr2^{-/-}$ mice (Supplementary Fig. S3). In contrast, RT-qPCR and western blotting revealed that the mRNA and protein levels of NLRP3, GSDMD and Caspase-1 in the colon tissue of $Mdr2^{-/-}$ mice were increased following PS-MPs exposure. Conversely, MCC950 treatment (a NLRP3 inhibitor) significantly ameliorated these changes (Fig. 3A–B). Additionally, MCC950 treatment significantly alleviated the PS-MP-induced increase in the mRNA levels of TNF- α , IL-1 β and IL-6 in

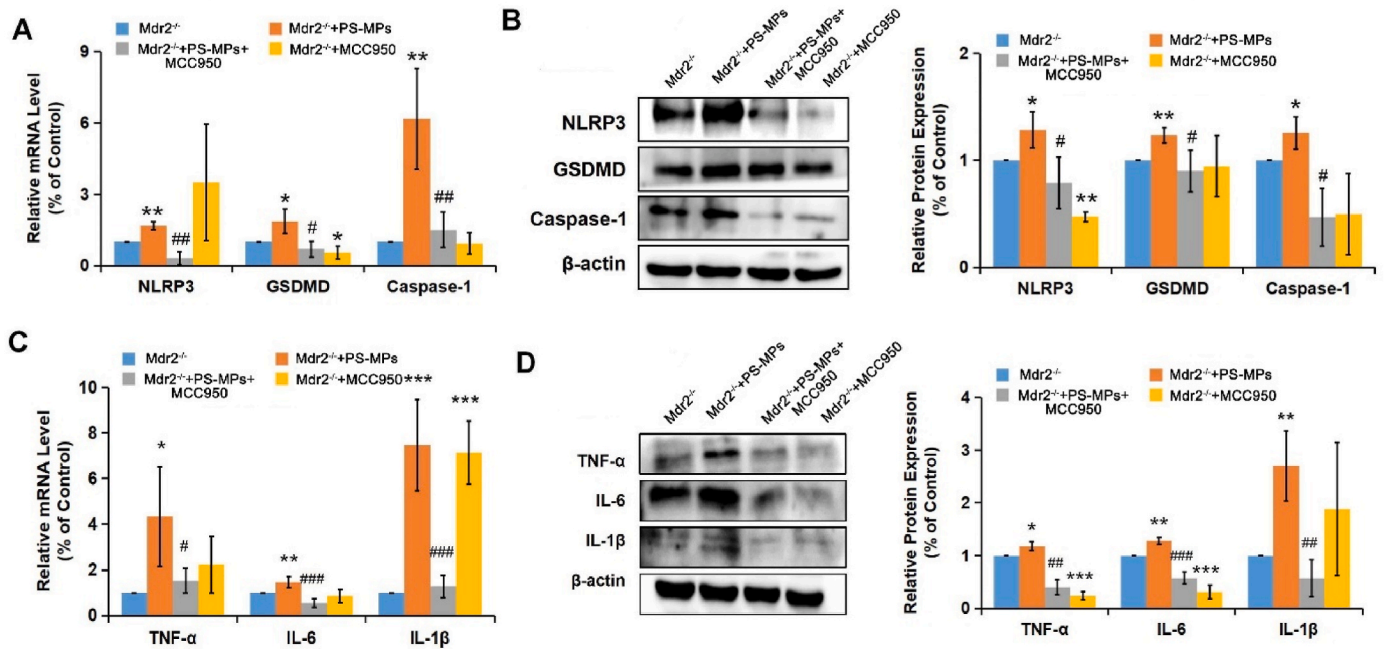
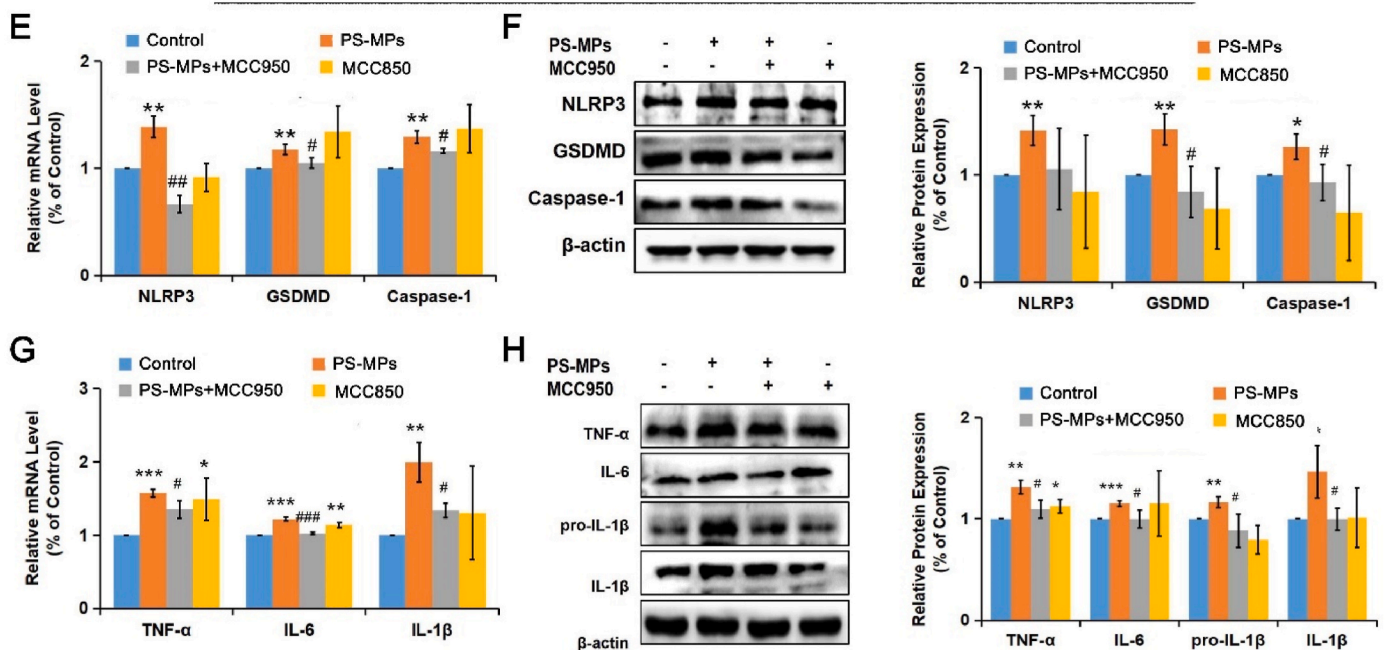
Colon tissue (*In vivo*)CaCO₂ cells (*In vitro*)

Fig. 3. NLRP3 ablation protected against PS-MPs-induced colon inflammation in *Mdr2*^{-/-} mice and Caco2 cells. *Mdr2*^{-/-} mice and Caco2 cells were subjected to MCC950 treatment and PS-MP exposure. (A) The relative mRNA and (B) protein expression levels of NLRP3, GSDMD and Caspase-1 in colon tissue were determined via qRT-PCR and western blotting. (C) The mRNA and (D) protein expression levels of TNF-α, IL-6 and IL-1β in colon tissue were determined by qRT-PCR and western blotting. (E) The mRNA and (F) protein expression levels of NLRP3, GSDMD and Caspase-1 in Caco-2 cells were determined by qRT-PCR and western blotting. (G) The mRNA levels of TNF-α, IL-6 and IL-1β in Caco-2 cells were determined via qRT-PCR and quantified. (D) The protein expression levels of TNF-α, IL-6, pro-IL-1β and IL-1β in Caco2 cells were determined by western blotting and quantified. *n* = 10 mice per group. Significant correlations are represented by **P* < 0.05, ***P* < 0.01, and ****P* < 0.01 vs. the *Mdr2*^{-/-} group or control group; #*P* < 0.05, ##*P* < 0.01, and ###*P* < 0.001 vs. the *Mdr2*^{-/-} + PS-MP group.

Mdr2^{-/-} mice (Fig. 3C–D). Consistent with the *in vivo* results, PS-MP exposure significantly increased the protein and mRNA levels of NLRP3, GSDMD and Caspase-1 in Caco-2 cells, and these effects were ameliorated by MCC950 treatment (Fig. 3E–F). In addition, the results revealed that the upregulated TNF-α, IL-1β, and IL-6 mRNA levels induced by PS-MPs were suppressed by MCC950 treatment in Caco-2

cells (Fig. 3G–H). Combining the above results, we propose that the ability of PS-MPs to aggravate CLD course development is attributed to NLRP3-driven inflammasome activation.

3.4. Inhibition of NLRP3 relieves PS-MPs-induced colon injury in *Mdr2*^{-/-} mice

The *in vivo* results indicated that MCC950 treatment significantly alleviated the PS-MPs-induced shortening of colon length in *Mdr2*^{-/-} mice (Fig. 4A–B). Furthermore, H&E staining of colon sections revealed that the layers of colon tissue in the PS-MPs + MCC950 group presented clear structures, an intact mucosal epithelium and scattered lymphocyte infiltration in the stroma without obvious inflammatory cell infiltration (blue arrow), which was significantly different from the severe injury observed in the PS-MPs group (Fig. 4C–D). Furthermore, AB-PAS staining analysis of colon sections revealed that compared with *Mdr2*^{-/-} mice, *Mdr2*^{-/-} mice exposed to PS-MPs presented decreased goblet cell density (green arrow) and mucus coverage (black arrow). In contrast, MCC950 pretreatment reversed the decreases in goblet cell density and mucus coverage in *Mdr2*^{-/-} mice after PS-MPs exposure (Fig. 4E–F). Therefore, our results suggest that inhibiting NLRP3 activation is a prerequisite for alleviating PS-MPs-induced colon damage in *Mdr2*^{-/-} mice.

3.5. The NLRP3 inflammasome is involved in PS-MPs-induced damage to the integrity of the colonic barrier in *Mdr2*^{-/-} mice and Caco-2 cells

To delineate the biological importance of NLRP3 in PS-MPs-induced damage to colon barrier integrity, we analysed the effects of NLRP3 inhibition on PS-MPs-induced intestinal barrier damage. The western blot results revealed that MCC950 treatment significantly improved the PS-MP-induced reduction in the expression of the barrier integrity ZO-1, Occludin and Claudin-1 in *Mdr2*^{-/-} mice (Fig. 5A). On the other hand, immunofluorescence analysis revealed that PS-MPs exposure significantly decreased the expression of ZO-1 and resulted in clear boundary injury, with gaps or even disappearance in Caco-2 cells. In contrast, MCC950 treatment notably reversed the reduction in ZO-1 expression in Caco-2 cells and improved the intestinal barrier integrity damage induced by PS-MPs (Fig. 5B–C). Consistent with the *in vivo* results, the RT-qPCR and western blotting results also demonstrated that MCC950 treatment improved the PS-MPs-induced reduction in the protein and mRNA levels of ZO-1, Occludin and Claudin-1 in Caco-2 cells (Fig. 5D–E). These results suggest that PS-MPs-induced intestinal barrier integrity damage results from the activation of the NLRP3 inflammasome.

3.6. NLRP3 inflammasome activation is involved in the PS-MPs-induced exacerbation of CLD

Next, we sought to clarify whether NLRP3 inflammasome inhibition alleviates PS-MPs-induced liver injury in *Mdr2*^{-/-} mice. In this study, the results showed that *Mdr2*^{-/-} mice exposed to PS-MPs presented significant weight loss from day 7, 14, 21 and 28 days compared with *Mdr2*^{-/-} mice. In contrast, MCC950 treatment improved PS-MPs-induced weight loss in *Mdr2*^{-/-} mice (Fig. 6A). Meanwhile, we also found that MCC950 treatment ameliorated the PS-MPs-induced decrease in the liver coefficient of *Mdr2*^{-/-} mice (Fig. 6B). Furthermore, MCC950 treatment significantly alleviated the PS-MPs-induced increases in ALT, AST and γ -GGT levels in the serum of *Mdr2*^{-/-} mice but did not affect ALP levels (Fig. 6C–D). Moreover, H&E staining revealed that MCC950 treatment significantly alleviated PS-MPs-induced lymphocytic infiltration, mild granular degeneration of hepatocytes, hepatocyte necrosis and fragmentation of karyopyknosis in *Mdr2*^{-/-} mice (Fig. 6E–F). In addition, Masson staining revealed that MCC950 treatment significantly suppressed PS-MPs-induced collagen deposition in *Mdr2*^{-/-} mice (Fig. 6G–H). These findings suggest that PS-MPs aggravated the course of CLD resulting from NLRP3 inflammasome activation.

3.7. Spearman correlation coefficient in *Mdr2*^{-/-} mice exposed to PS-MPs

Through correlation analysis to further verify the connection among intestinal barrier integrity genes (ZO-1, Occludin and Claudin-1), colonic proinflammatory genes and serum hepatobiliary damage index (ALT, AST, ALP, γ -GGT), it was found that there was a significant positive correlation between ALP/ γ -GGT levels and intestinal barrier integrity genes, as well as a significant positive correlation with IL-1 β after PS-MPs exposure (Fig. 7). However, we found no significant correlation between colonic NLRP3 gene and serum hepatobiliary damage index. In contrast, there was a significant positive correlation between NLRP3 mRNA levels and intestinal barrier integrity genes (Fig. 7). This proved that PS-MPs exposure induce intestinal barrier integrity damage by stimulating NLRP3 pathway, and subsequently triggering the hepatobiliary damage, probably leading to CLD.

4. Discussion

These results showed that PS-MPs exposure aggravated cholestatic hepatic fibrosis progression. Additionally, our data demonstrated that PS-MPs exposure resulted in intestinal barrier integrity damage via regulation of the NLRP3-driven inflammasome in *Mdr2*^{-/-} mice. Furthermore, treatment with an NLRP3 inhibitor attenuated the progression of PS-MPs-induced cholestatic hepatic fibrosis. Overall, the insights gained from this study will improve our understanding of PS-MPs-induced hepatobiliary injury and will be crucial for advancing future clinical research on hepatobiliary protection.

Previous mouse experiments have confirmed that PS-MPs can increase the sensitivity of the liver to the toxic effects of cyclophosphamide, thereby triggering more severe histological damage (Wen et al., 2022). Moreover, a previous study demonstrated that PS-MPs can cause secondary liver injury in mice with colitis (Luo et al., 2022). Consistently, the present study confirmed that PS-MPs exposure exacerbates liver connective tissue hyperplasia, increases the severity of fibrosis, and increases inflammation in CLD mice. In terms of anatomy and physiology, the liver and intestine are connected by the portal vein and bile duct. Most of the blood in the intestine flows into the liver through the portal vein, affecting liver homeostasis. On the other hand, the imbalance of the intestinal microecology contributes to liver damage or dysfunction by affecting hepatic metabolic pathways or resident immune cells (Xu et al., 2024). Furthermore, gut barrier dysfunction induces the transfer of bacteria or bacterial products, thus facilitating the occurrence of liver fibrosis, cirrhosis, and other liver diseases (Shi et al., 2022b). Previous data suggest that remodelling of the gut flora by faecal microbiota transplantation could improve the health status of patients with CLD (Allegretti et al., 2019). Therefore, the further aggravation of CLD development by PS-MPs may be attributed to an intestinal microecological imbalance.

The gut serves as the body's primary defense against nonspecific infections. It functions as a mechanical, immune, and chemical barrier. The intestinal epithelial barrier is formed by intestinal epithelial cells and the mucus layer under physiological conditions. The existence of intestinal barrier integrity is crucial for maintaining the integrity of the intestinal epithelium. The mechanisms of intestinal barrier dysfunction are complex and diverse. Disruption of tight junctions increases gut permeability, allowing the translocation of bacteria and enterotoxins and causing liver dysfunction development (Lv et al., 2023; Shi et al., 2024). Here, our study confirmed that PS-MPs decreased the expression of ZO-1, Occludin and Claudin-1 and increased the expression of proinflammatory cytokines in the colons of *Mdr2*^{-/-} mice. In addition, AB-PAS staining revealed that PS-MPs decreased the number of colonic goblet cells, which is consistent with the commonly reported damaging effect of PS-MPs on the intestinal barrier (Jin et al., 2019). Consistently, previous studies have also demonstrated that the concentration of MPs in mice with intestinal immune imbalance increases, which may

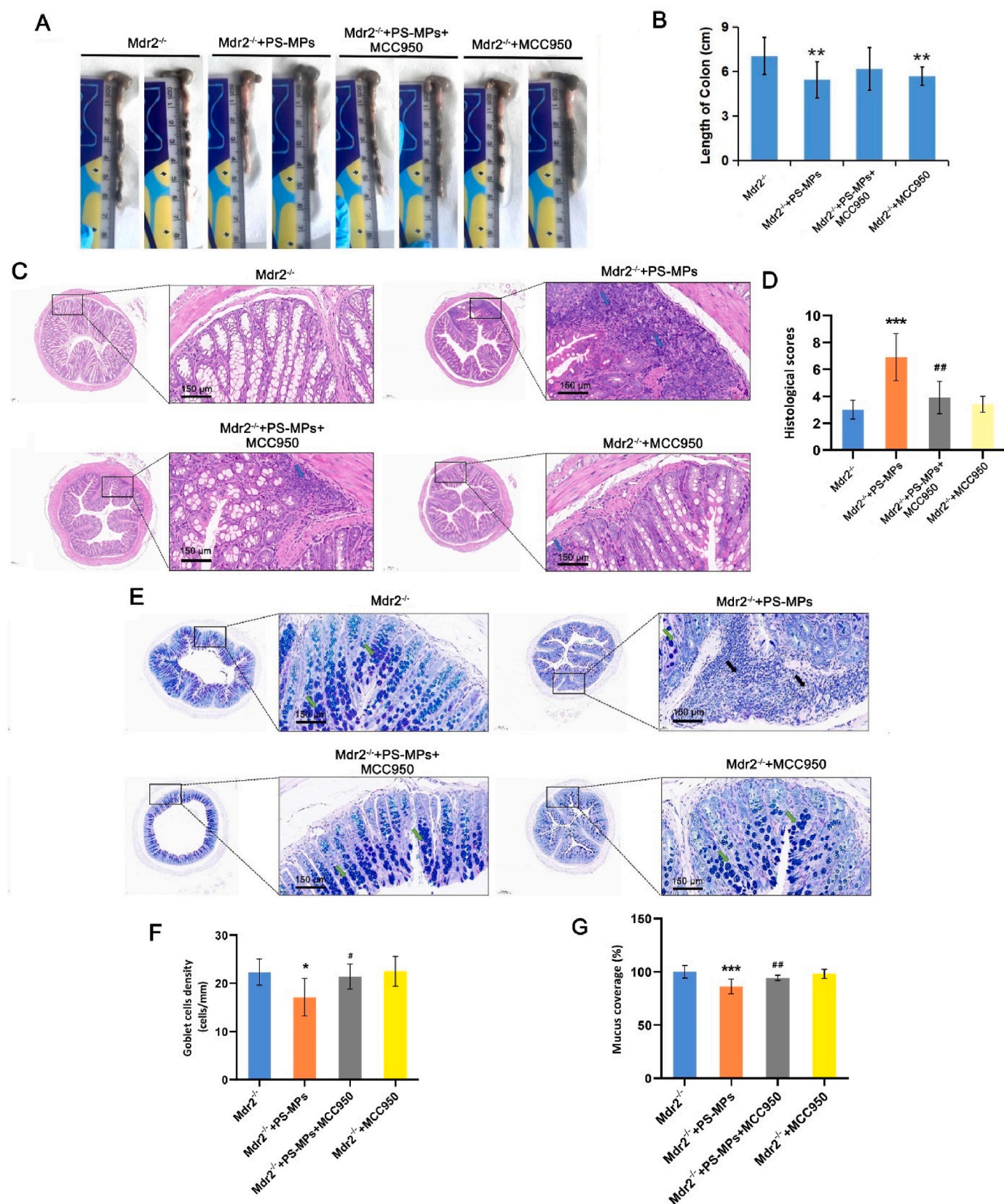


Fig. 4. NLRP3 ablation improved PS-MP-induced colon damage in *Mdr2*^{-/-} mice. *Mdr2*^{-/-} mice were subjected to MCC950 treatment and PS-MP exposure. (A–B) Colon length and morphology of *Mdr2*^{-/-} mice. (C) H&E staining of colon sections. Inflammatory cell infiltration is shown by blue arrows. (D) Representative colon histological scores (H&E). (E) AB–PAS staining of colon sections. Goblet cells are shown by green arrows, and the focal decreased number of lamina propria intestinal glands is shown by black arrows. (F) The corresponding goblet cell densities and (G) mucus coverage. The above results were obtained from at least three independent replicates. *n* = 10 mice per group. Significant correlations are represented by **P* < 0.05, ***P* < 0.01 vs. the *Mdr2*^{-/-} group; #*P* < 0.05, ##*P* < 0.01 vs. the *Mdr2*^{-/-} + PS-MPs group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

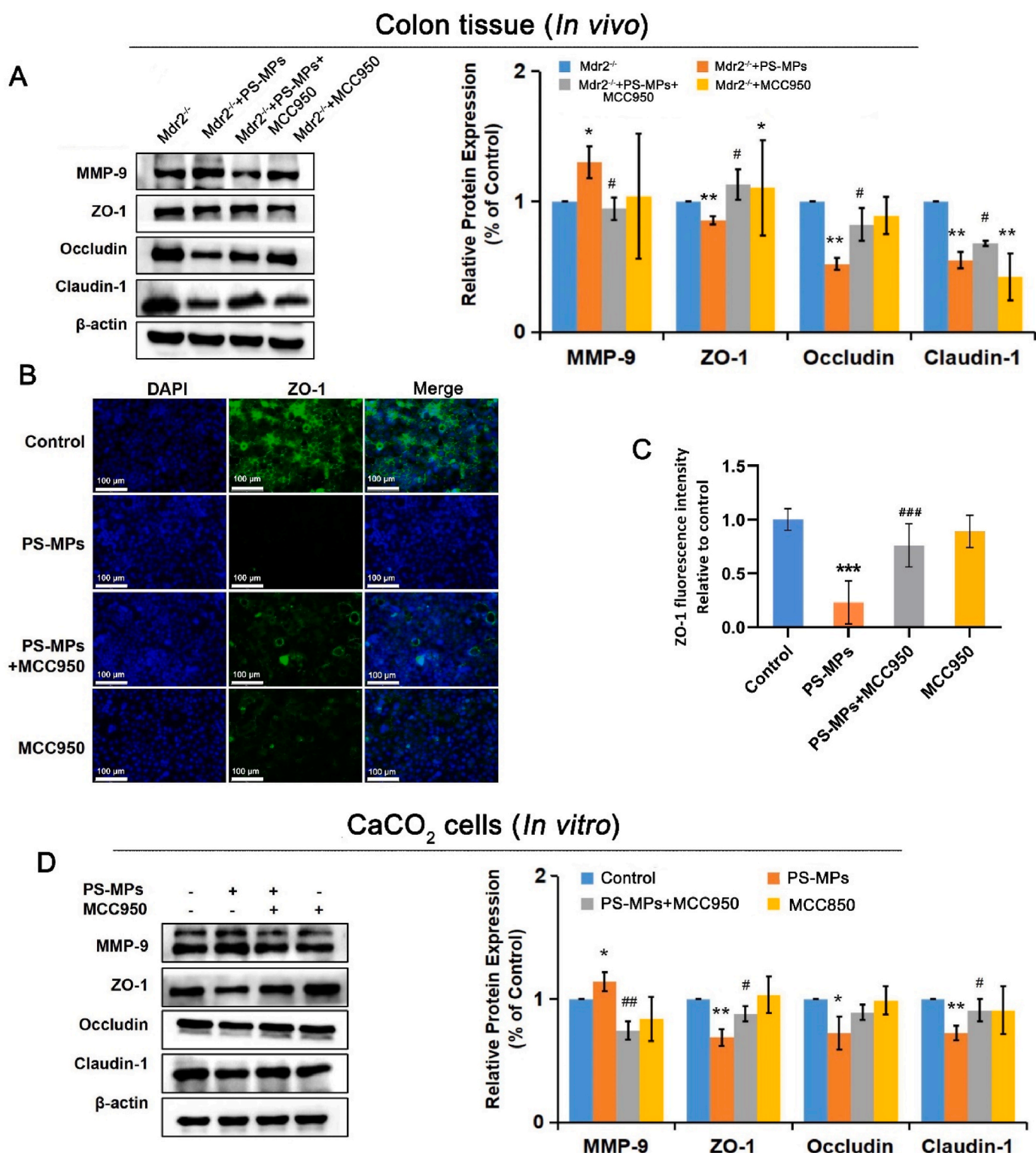


Fig. 5. NLRP3 ablation ameliorates PS-MP-induced damage to intestinal barrier integrity in Mdr2^{-/-} mice and Caco2 cells. (A) The protein levels of MMP-9, ZO-1, Occludin and Claudin-1 in colon tissue were determined by western blotting. (B) Immunofluorescence staining of ZO-1 in Caco-2 cells at the *in vitro* level (20 ×) by fluorescence microscopy. The blue fluorescence represents the nucleus, and the green fluorescence represents the ZO-1 protein. (C) Quantification of ZO-1 fluorescence intensity. (D) The protein expression of MMP-9, ZO-1, Occludin and Claudin-1 in Caco-2 cells was determined by western blotting. n = 10 mice per group. Significant correlations are represented by **P* < 0.05, ***P* < 0.01 vs. the Mdr2^{-/-} mouse group or control group; #*P* < 0.05, ###*P* < 0.01 vs. the PS-MP group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

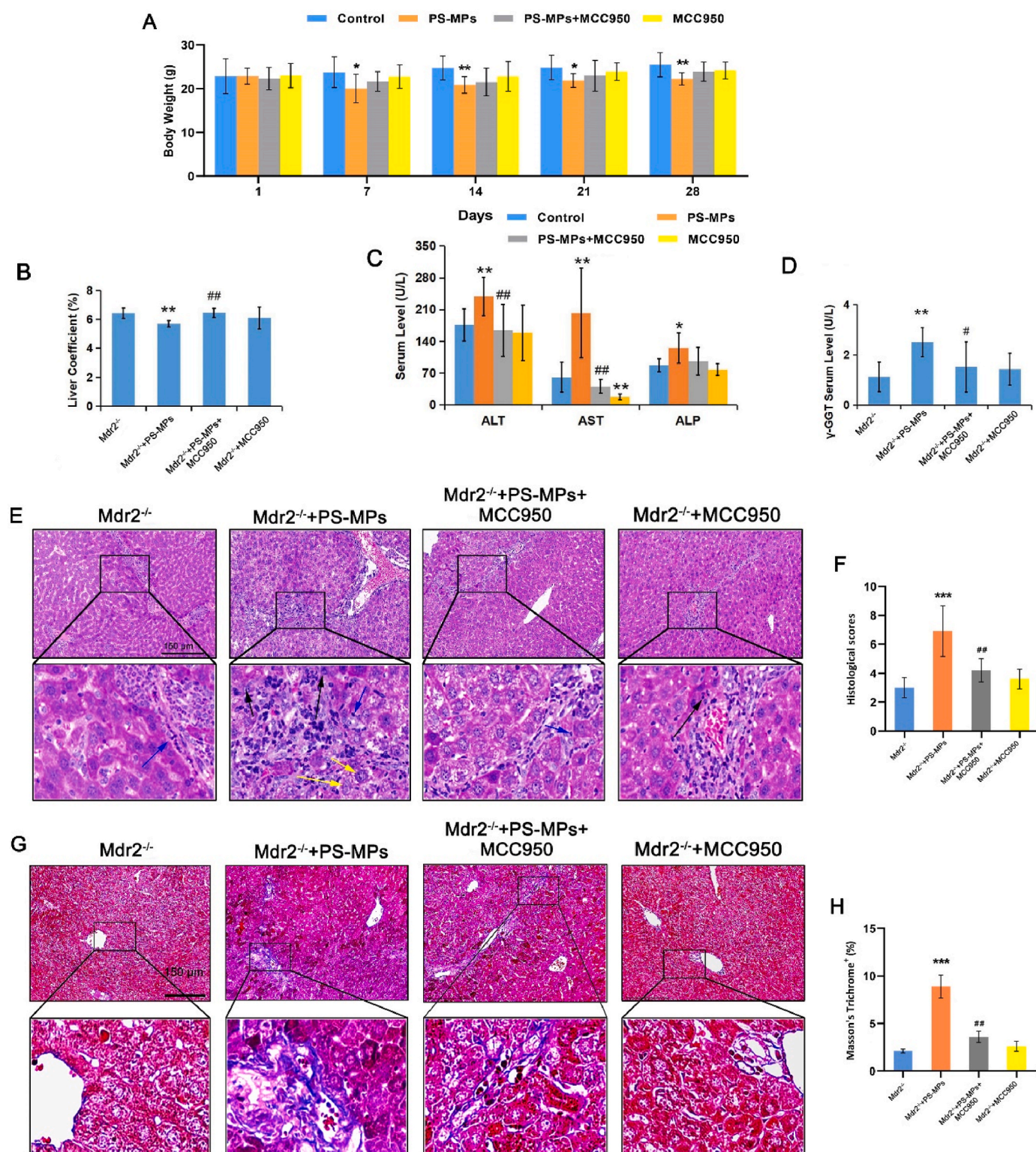


Fig. 6. NLRP3 ablation ameliorates PS-MP-induced exacerbation of hepatobiliary injury in Mdr2^{-/-} mice. (A) Body weights. (B) Liver coefficient. (C) Serum ALT, AST, and ALP levels. (D) Serum γ-GGT levels. (E) H&E staining of liver sections. More severe inflammatory cell infiltration is shown by the blue arrow, mild granular degeneration of hepatocytes is shown by the black arrow, and slight vacuolar degeneration of hepatocytes is shown by the yellow arrow. (F) Representative liver histological scores (H&E). (G) Masson staining of liver sections. (H) Quantification of Masson's trichrome staining. n = 10 mice per group. Significant correlations are represented by **P* < 0.05, ***P* < 0.01, ****P* < 0.001 vs. the Mdr2^{-/-} mouse group. #*P* < 0.05, ##*P* < 0.01 vs. the Mdr2^{-/-} + PS-MPs group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

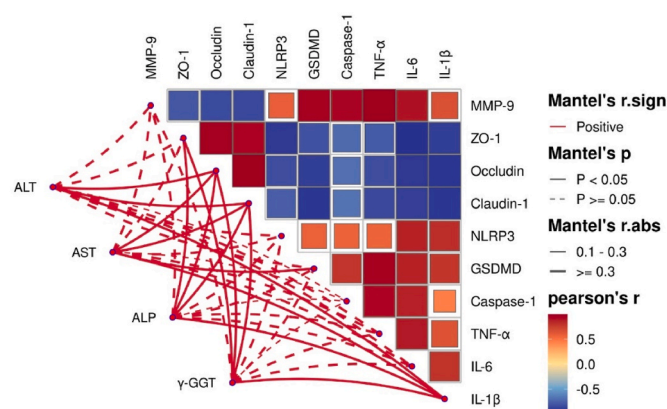


Fig. 7. The Spearman correlation coefficient between intestinal barrier integrity genes and liver function index in *Mdr2*^{-/-} mice exposed to PS-MPs.

contribute to increased intestinal toxicity (Liu et al., 2022). Moreover, PS-MPs have been demonstrated to trigger intestinal inflammation in zebrafish (Duan et al., 2024). Taken together, our results suggest that PS-MPs exposure can destroy the tight junction structure and cause severe damage to the intestinal barrier, with widespread inflammatory cellular infiltration, thereby contributing to liver toxicity.

Certainly, MPs can directly impair liver structure through intestinal absorption, as well as indirectly disrupt liver function via dysregulating the barrier of the intestinal epithelia. In this study, through correlation analysis to further verify the connection between ALP/γ-GGT levels and intestinal barrier integrity genes, as well as a significant positive correlation with IL-1β after PS-MPs exposure. Previous study has indicated that MPs affect the partial co-metabolism in the intestine and the liver, suggesting a link to the “gut-liver axis” (Wang et al., 2024). We therefore speculate that the association between the intestinal barrier damage and MPs-aggravated CLD also likely attribute to co-metabolism disorder (Fackelmann and Sommer, 2019). On the other hand, the intestinal barrier function is crucial for liver injury development because harmful bacteria and metabolites in the gut enter the blood through the weakness gut, subsequently reaching the liver, thus leading to liver disease. MPs have been demonstrated to alter the composition of gut microbiota, which will in turn lead to intestinal barrier dysfunction and metabolic disorder. Therefore, intestinal barrier injury drives intestinal flora to enter liver tissue, and immune-driven liver injury may be another mechanism by which PS-MPs exacerbates CLD. However, this may only be part of the mechanism of MPs hepatotoxicity, as the presence of MPs locally in the liver suggests that MPs may also contribute directly to liver injury.

The inflammasome is a crucial component of the innate immune response, and its signal transduction is vital in the intestinal immune process (Man, 2018). Several reports have demonstrated that the intake of PS-MPs is associated with the activation of the NLRP3 inflammasome in the body. For example, PS-MPs can penetrate the rat ovary and be assimilated by granulosa cells, leading to pyroptosis through the NLRP3/Caspase-1 signalling pathway (Hou et al., 2021). PS-MPs exposure causes cardiac injury via activation of the NF-κB-NLRP3-GSDMD pathway (Zhang et al., 2022). Moreover, PS-MPs exposure can exacerbate LPS-induced duodenal permeability injury and inflammation in mice by activating the ROS-driven NF-κB/NLRP3 pathway (He et al., 2022). There is strong evidence suggesting that NLRP3 contributes to the maintenance of intestinal homeostasis and the regulation of the intestinal flora (Xu et al., 2024). Consistently, our data revealed that PS-MPs exposure activated the NLRP3-driven inflammatory pyroptosis pathway in the colons of *Mdr2*^{-/-} mice. Furthermore, we found that the inhibition of NLRP3 inflammasome activation could alleviate PS-MP-induced intestinal barrier integrity damage and hepatic fibrosis in *Mdr2*^{-/-} mice. Pathological changes in intestinal epithelial cells and

damage to the intestinal mucosal barrier in the gastrointestinal system can reduce the body's ability to resist pathogenic bacteria, leading to gastrointestinal diseases. Previous results suggested that the accumulation of PS-MPs in mammalian organs was positively correlated with exposure duration (Xu et al., 2023). On the basis of these observations, we propose that pyroptosis in the colon induced by PS-MPs may be the main cause of the development of CLD. Additionally, we propose that PS-MP-induced colonic pyroptosis may be an important environmental factor in the occurrence or pathogenesis of CLD.

Our study had some limitations. First, we did not quantify the distribution of tissues and organs in the whole body of *Mdr2*^{-/-} mice after PS-MPs exposure. Therefore, we could not determine the distribution rule of liver and intestine in *Mdr2*^{-/-} mice after PS-MPs exposure. Second, although we assessed that obstructing intestinal barrier damage could alleviate the pathological progression of MPs exacerbating CLD, the drugs used were not highly specific to the gut. Third, although correlation analysis confirmed a positive association between intestinal barrier integrity and liver function index, we did not conduct relevant intervention experiments to determine causation.

5. Conclusion

Our findings suggest that disruption of intestinal barrier integrity by PS-MPs can be one of the factors aggravating the course of CLD. Additionally, PS-MPs exposure resulted in intestinal barrier integrity damage via regulation of the NLRP3-driven inflammasome in *Mdr2*^{-/-} mice. In addition, the inhibition of NLRP3 activation has a protective effect on the progression of *Mdr2*^{-/-} mice after PS-MPs exposure. The results of this study offer new insights into the mechanisms underlying PS-MPs-induced hepatobiliary injury and will be crucial for advancing future clinical research on hepatobiliary protection.

CRediT authorship contribution statement

Siwen Li: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Yu Yuan:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Yuan Zeng:** Investigation, Formal analysis. **Peng Fang:** Resources, Formal analysis. **Junchang Wang:** Resources, Formal analysis. **Yi Liu:** Resources, Formal analysis. **Fang Xiao:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2025.125753>.

Data availability

No data was used for the research described in the article.

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